/* 2) Write a C program that contains a string (char pointer) with a value "Helloworld". The program should AND or XOR each character in this string with 127 and display the result. */

```
/* ===== 1) XOR ===== */

#include <stdio.h>
#include <string.h>

int main()
{
   char String[] = "Helloworld";
char cipher[100];     char
plain[100];     int len =
strlen(String);

   for (int i = 0; i < len; i++)
   {
      cipher[i] = String[i] ^ 127;
   }     cipher[len] = '\0';
printf("Cipher text: %s\n", cipher);

   for (int i = 0; i < len; i++)
   {        plain[i] = cipher[i] ^
127;
   }     plain[len] = '\0';
printf("Plain text: %s\n", plain);

   return 0;
}

/* Output:-
Cipher text: 7<non-printable characters>
Plain text: Helloworld
*/


/* ===== 2) AND ===== */

#include <stdio.h>
#include <string.h>

int main()
{
   char String[] = "Helloworld";
```

```
   char cipher[100];
char plain[100];     int len
= strlen(String);

   for (int i = 0; i < len; i++)
   {
      cipher[i] = String[i] & 127;
   }     cipher[len] = '\0';
printf("Cipher text: %s\n", cipher);

   for (int i = 0; i < len; i++)
   {
      plain[i] = cipher[i] & 127;
   }     plain[len] = '\0';
printf("Plain text: %s\n", plain);

   return 0;
}

/* Output:-
Cipher text: Helloworld
Plain text: Helloworld
*/
```

3) Calculate the message digest of text using MD5 Algorithm.

```
import java.security.*;
public class MD5 {
public static void main(String[] a) {
// TODO code application logic here
try {
MessageDigest md = MessageDigest.getInstance("MD5");
System.out.println("Message  digest object info: ");
System.out.println(" Algorithm = " +md.getAlgorithm());
System.out.println(" Provider = " +md.getProvider());
System.out.println(" ToString = " +md.toString());
String input = "";
md.update(input.getBytes());
byte[] output = md.digest();
System.out.println();
System.out.println("MD5(\""+input+"\") = " +bytesToHex(output));
input = "abc";
md.update(input.getBytes());
output = md.digest();
System.out.println();
System.out.println("MD5(\""+input+"\") = " +bytesToHex(output));
input = "abcdefghijklmnopqrstuvwxyz";
md.update(input.getBytes());
output = md.digest();
System.out.println();
System.out.println("MD5(\"" +input+"\") = "
+bytesToHex(output)); System.out.println("");
}
catch (Exception e) {
System.out.println("Exception:  " +e); }
```

```
}
public static String bytesToHex(byte[] b) {
char hexDigit[] = {'0', '1', '2', '3', '4', '5', '6', '7', '8', '9', 'A', 'B', 'C', 'D', 'E', 'F' };
StringBuffer buf = new StringBuffer();
for (int j=0; j<b.length; j++) {
buf.append(hexDigit[(b[j] >> 4) & 0x0f]);
buf.append(hexDigit[b[j] & 0x0f]); }
return buf.toString(); } }
```

Output

```
Message digest object info:
 Algorithm = MD5
 Provider = SUN version 17
 ToString = MD5 Message Digest from SUN, <initialized>

MD5("") = D41D8CD98F00B204E9800998ECF8427E

MD5("abc") = 900150983CD24FB0D6963F7D28E17F72

MD5("abcdefghijklmnopqrstuvwxyz") = C3FCD3D76192E4007DFB496CCA67E13B
```

4) Write a program to calculate SHA-512 value.

```
import java.math.BigInteger;
import java.security.MessageDigest;
import java.security.NoSuchAlgorithmException;
import java.util.*;

public class SHA512 {
   public static void main(String args[]) throws NoSuchAlgorithmException {
      Scanner sc = new Scanner(System.in);
      System.out.println("Enter input String(Plain text Message): ");
      String str = sc.next();
      String hashcode;
      MessageDigest md;

      try {
         md = MessageDigest.getInstance("SHA-512");
         byte[] messageDigest = md.digest(str.getBytes());
         BigInteger no = new BigInteger(1, messageDigest);
         hashcode = no.toString(16);

         while (hashcode.length() < 32) {
            hashcode = "0" + hashcode;
         }
      } catch (NoSuchAlgorithmException e) {
         throw new RuntimeException(e);
      }

      System.out.println("HashCode Generated by SHA-512 for: ");
      System.out.println("\n" + str + " is : " + hashcode);
      System.out.println("Message digest object info: ");
      System.out.println(" Algorithm = " + md.getAlgorithm());
      System.out.println(" Provider = " + md.getProvider());
      System.out.println(" ToString = " + md.toString());
   }
}
```

Output:-

```
Enter input String(Plain text Message):
hello
HashCode Generated by SHA-512 for:
hello is :
9b71d224bd62f3785d96d46ad3ea3d73319bfbc2890caadae2dff72519673ca72323c3d99ba5
c11d7c7acc6e14b8c5da0c4663475c2e5c3adef46f73bcdec043 Message digest object info:
Algorithm = SHA-512
 Provider = SUN version 17
 ToString = SHA-512 Message Digest from SUN, <initialized>
```